



University of Moratuwa

Faculty of Engineering

Department of Electronic and Telecommunication
Engineering

EN2111

Electronic Engineering Design

Tri-mode Capable Audio Power Amplifier

Approach, Design & Functionality

July 26, 2026

Group 19

230458P	H.H. Palihena
230564L	S.M.R.P. Samarasekara
230566U	S.M.R.R. Samarasinghe
230613M	M.N.N. Shehan

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1 Introduction

A tri-mode capable audio power amplifier is a power amplifier that can drive two speakers in three distinct configurations. This allows the system to adapt to different user environments and power needs without altering the internal components.

- **Mono Mode** – The Left and Right channels from the stereo input are summed and amplified to drive both speakers with the same audio signal.
- **Stereo Mode** – The Left and Right channels from the stereo input are individually amplified. Each channel drives only one speaker.
- **Bridge Mode** – Achieves a theoretical four-fold power output by driving only one speaker utilizing both power amplifiers.

2 System Architecture & Approach

2.1 Tri-Mode Routing Block Diagrams

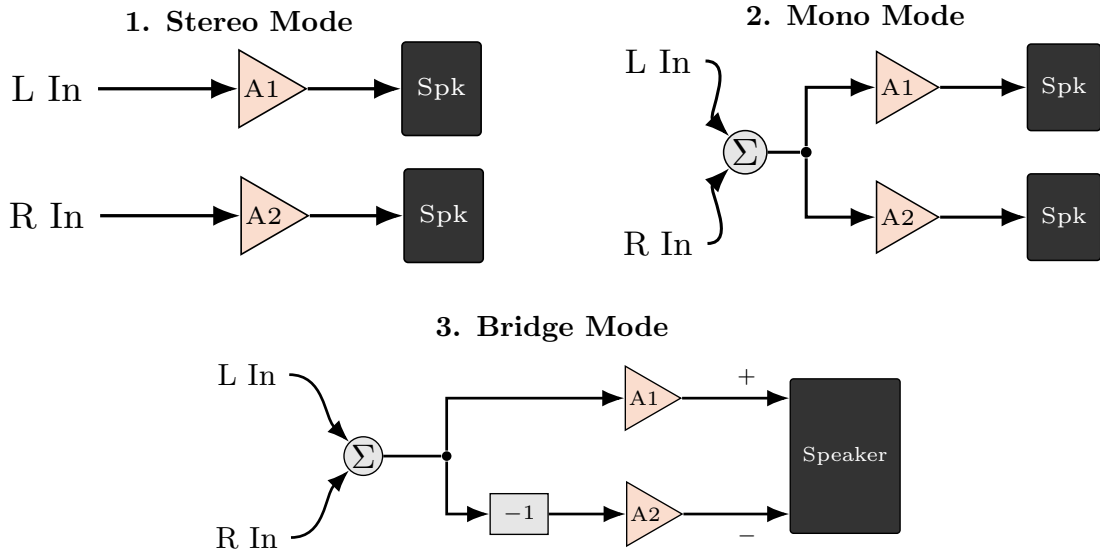


Figure 1: Signal routing paths for Stereo, Mono, and Bridge configurations.

2.2 Class-AB Power Amplifier

The core amplifier is designed using a discrete Class-AB push-pull topology.

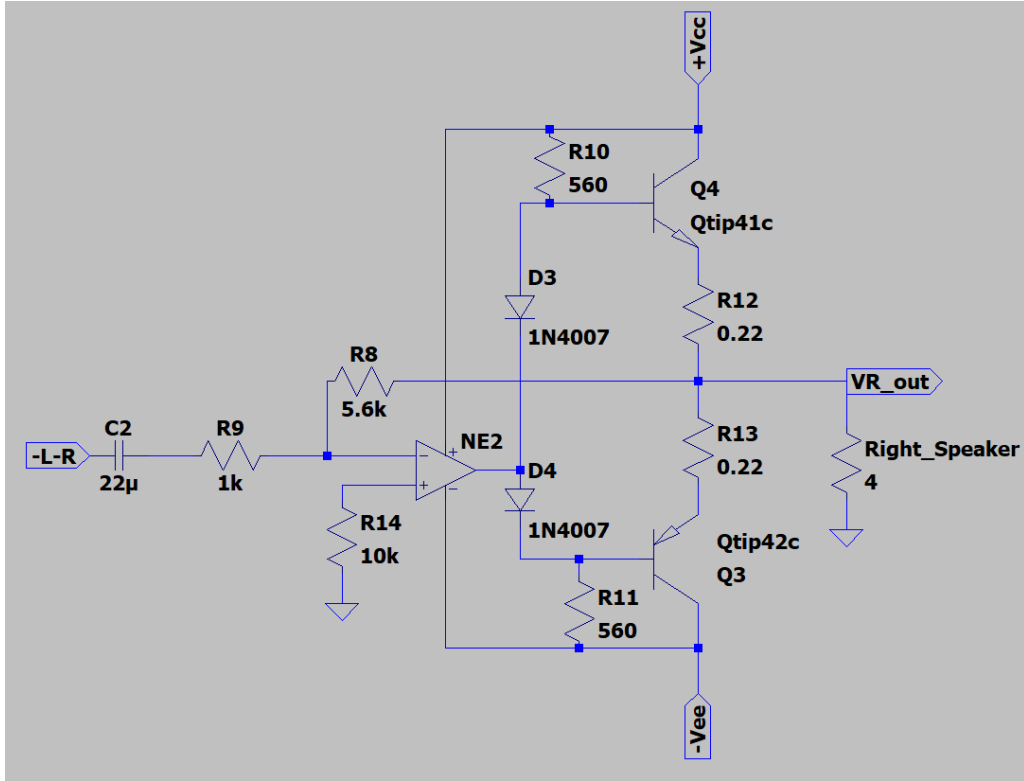


Figure 2: Schematic of the Class-AB Power Amplifier stage.

Circuit Operation & Structure:

- **Voltage Stage:** The OpAmp provides the required voltage gain at the output. This is adjusted by varying the feedback resistor network. The OpAmp feedback system compares the input and output voltages to maintain the set gain.
- **Current Stage:** The complementary pair of power BJTs provides the heavy current needed to drive the speaker directly from the power supply rails. The OpAmp output current feeds into the bases of these two BJTs.
- **Crossover Distortion Elimination:** Two PN junction diodes are placed in series to ensure both BJTs remain slightly forward-biased in the active region at all times, overcoming standard Class-B crossover distortion.
- **Thermal Stability:** 0.22Ω emitter degeneration resistors protect the circuit from thermal runaway. A 560Ω resistor across the collector and base ensures the biasing diodes remain sufficiently forward biased.

2.3 Left and Right Channel Summer & Inverter

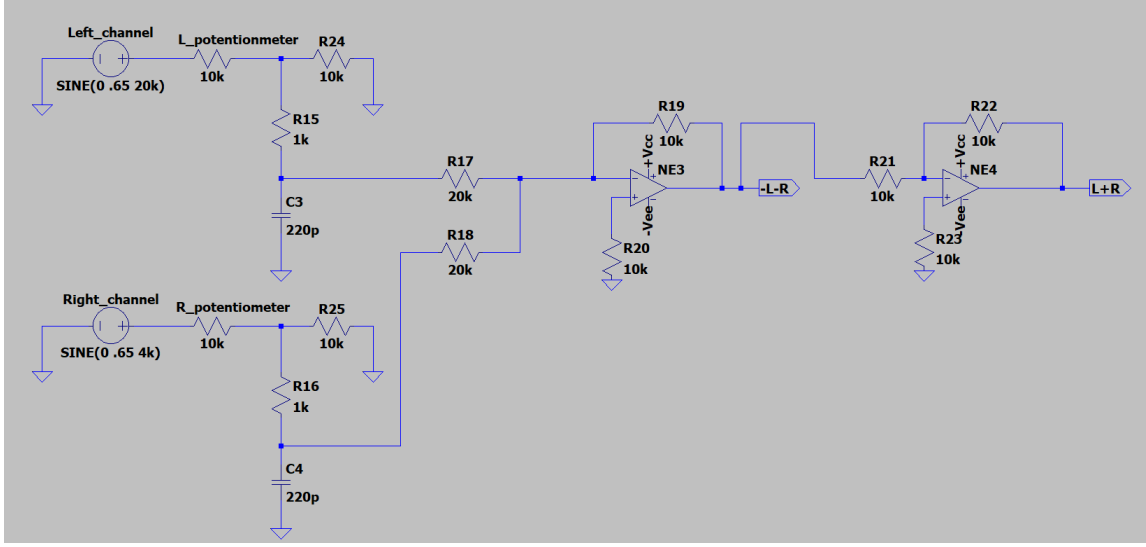


Figure 3: Schematic of the summing and inverting stage.

Circuit Operation:

- The two channel voltages are summed and inverted with a predefined gain of **0.5**. This attenuation ensures the combined output waveform will not clip against the power rails during peak amplitudes.
- The resulting signal is inverted again in a secondary stage to restore the correct phase, providing a pure (L+R) signal for Mono and Bridge modes.

3 Component Selection

3.1 Operational Amplifier (NE5532)

Requirements:

- Should be a **low noise** OpAmp suitable for high-fidelity audio.
- Must have a sufficient slew rate for 20kHz operation. The minimum required slew rate is calculated as:

$$SR \geq |V_{pp} \cdot j\omega e^{j\omega t}| = 7 \times 2\pi \times 20 \times 10^3 = 0.879 \text{ V } / \mu\text{s}$$

- A rail-to-rail output across the operational bandwidth is preferred.
- Must be fully operational under a $\pm 15V$ dual power supply.

Selection Justification: The **NE5532** was selected as it is an industry-standard audio-grade OpAmp featuring very low noise. It far exceeds the minimum slew rate requirement with a rating of $9 \text{ V}/\mu\text{s}$, and safely operates within a supply voltage range of $\pm 3V$ to $\pm 20V$.

3.2 Power Transistors (TIP41 & TIP42)

Requirements:

- Must be a **complementary pair** of NPN and PNP BJTs to ensure perfectly symmetrical performance in both positive and negative output half-cycles.
- Must support a continuous current of at least $1.5A$ and a maximum power dissipation rating of $> 3W$.
- Must be packaged in a format that easily mounts to an aluminum heat sink.

Selection Justification: The **TIP41 (NPN)** and **TIP42 (PNP)** were chosen for their matched characteristics. They comfortably supply a continuous $6A$ current and feature a high $65W$ power rating. Their TO-220 package provides excellent thermal dissipation capabilities when mounted to external heatsinks.

3.3 Biasing Diodes (1N4007)

Requirements:

- Must have a combined forward voltage drop of approximately $2V$ to safely bring the power transistors into their active state ($V_{BE(on)} \approx 2V$).
- Must support a continuous current of at least $1A$.

Selection Justification: Standard **1N4007** silicon rectifiers were selected. Connecting two in series provides a stable voltage drop (roughly $1.1V$ per diode under standard bias, totaling $\approx 2.2V$), perfectly balancing the base-emitter junctions of the TIP41/42 pair. They easily handle the required forward current and are readily available.

4 Simulation & Functionality (LTSpice)

4.1 Gain Verification

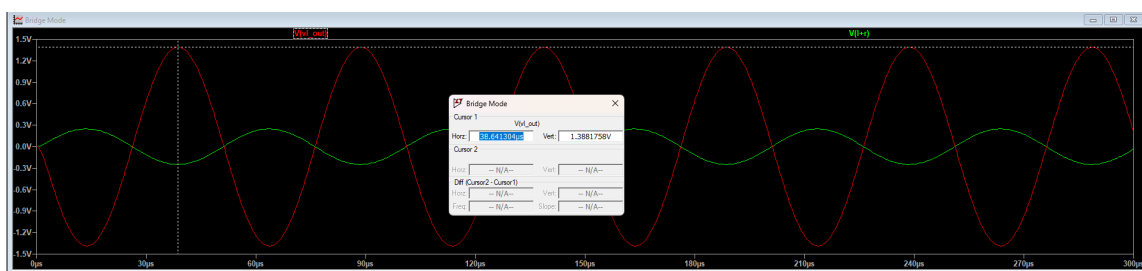


Figure 4: Analysis showing voltage gain.

Simulations verified that the OpAmp feedback circuit accurately provides the voltage gain set by the resistor pairs. The simulation above demonstrates a gain of ≈ 5.6 , achieved using a $5.6k\Omega$ and $1k\Omega$ resistor configuration.

4.2 Elimination of Crossover Distortion

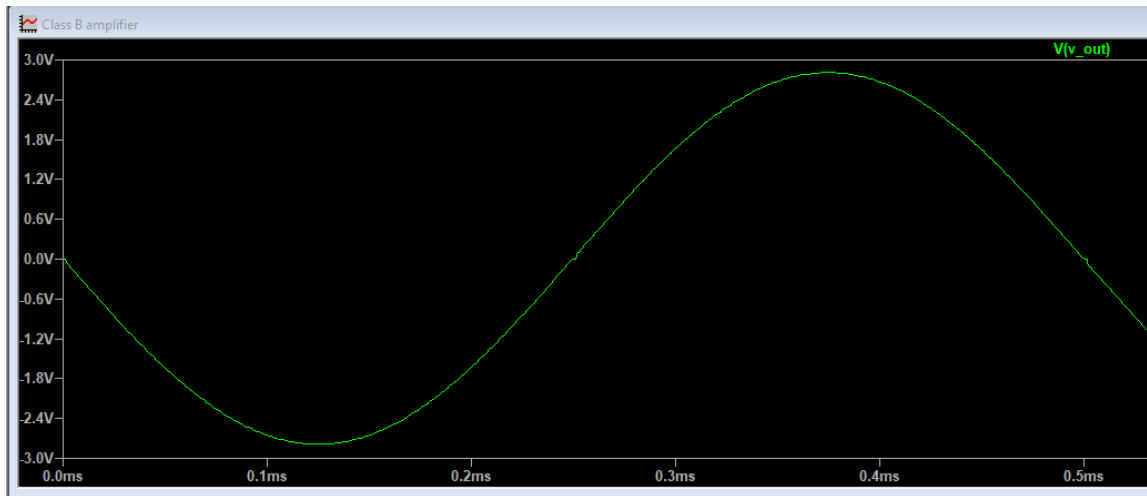


Figure 5: Simulated waveform illustrating crossover distortion when biasing diodes are omitted.

The design was tested without the diode network to observe standard Class-B behavior. It was subsequently verified that placing the biasing diodes within eliminated the crossover distortion present in a class B amplifier.

5 Testing & Analysis

All hardware testing and quantitative data analysis were performed using a 4Ω load.

5.1 Input vs. Output Signal Comparison

To verify signal integrity and voltage amplification, the input signal was directly compared with the output signal using an oscilloscope.

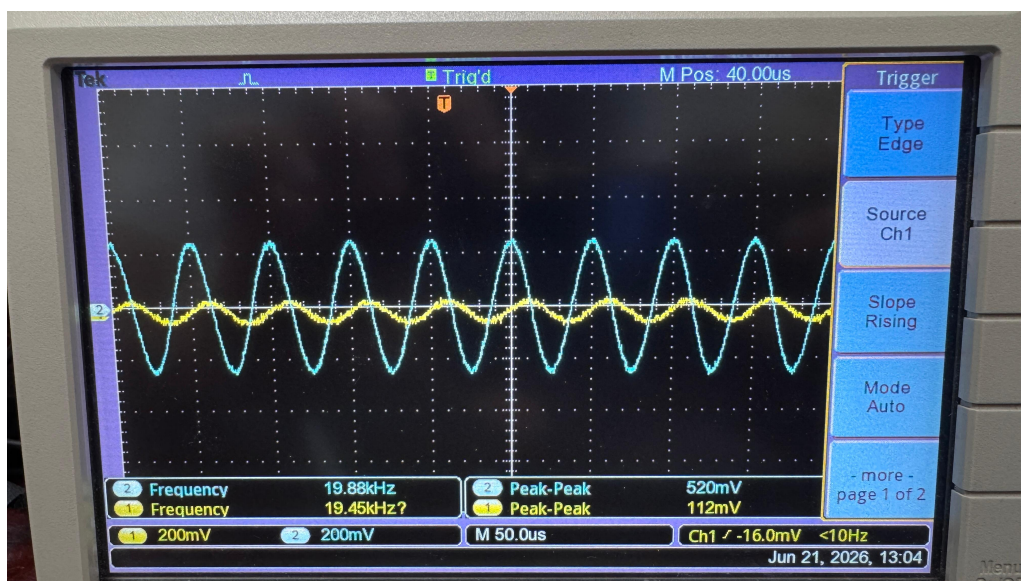


Figure 6: Oscilloscope trace comparing the input (yellow) and output (blue) signals at $\approx 20\text{kHz}$.

As shown in Figure 6, the amplifier successfully tracks and amplifies a high-frequency ($\approx 20\text{kHz}$) sine wave without introducing noticeable clipping or significant phase shift. The input signal was measured at 112 mV peak-to-peak, while the output signal reached 520 mV. This yields a practical voltage gain of approximately 4.64.

5.2 Power and Efficiency Metrics

5.2.1 Output Power (Stereo/Mono Mode)

The maximum output power (P_{out}) delivered to the load is calculated from the measured Peak-to-Peak voltage (V_{pp}) across the 4Ω load (R_L).

$$P_{out} = \frac{V_{rms}^2}{R_L} = \frac{\left(\frac{V_{pp}}{2\sqrt{2}}\right)^2}{R_L} = \frac{V_{pp}^2}{8 \cdot R_L} \quad (1)$$

Table 1: Measured Output Power in Stereo/Mono Mode ($R_L = 4\Omega$)

Supply Voltage ($\pm V_{CC}$)	Max Output V_{pp} (V)	Calculated P_{out} (W)
10 V	9.6	2.88
11 V	10.2	3.25
12 V	11.8	4.35
13 V	12.2	4.65
14 V	12.6	4.96
15 V	12.8	5.12
16 V	13.8	5.95
17 V	14.2	6.30
18 V	14.6	6.66
19 V	15.4	7.41
20 V	15.6	7.61

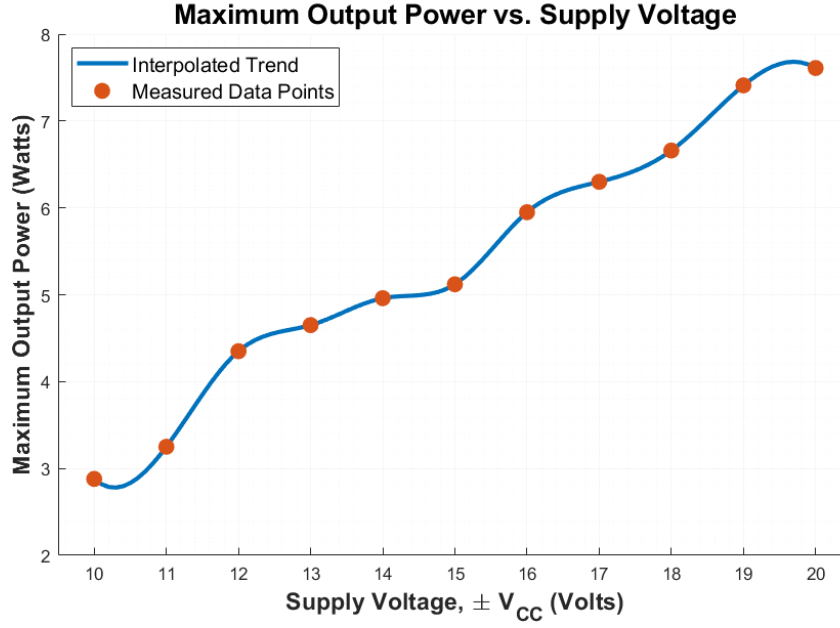


Figure 7: Max Output Power vs Supply Voltage (Stereo/Mono)

5.2.2 Output Power (Bridge Mode)

In Bridge-Tied Load configuration, the speaker is driven by both amplifiers. The output power is calculated using the same formula: $P_{out} = \frac{V_{pp}^2}{8 \cdot R_L}$.

Table 2: Measured Output Power in Bridge Mode ($R_L = 4\Omega$)

Supply Voltage ($\pm V_{CC}$)	Max Output V_{pp} (V)	Calculated P_{out} (W)
10 V	18.4	10.58
11 V	20.4	13.00
12 V	23.2	16.82
13 V	24.4	18.61
14 V	25.2	19.85
15 V	27.0	22.78
16 V	27.6	23.81
17 V	28.8	25.92
18 V	29.4	27.01
19 V	30.0	28.13
20 V	30.2	28.50

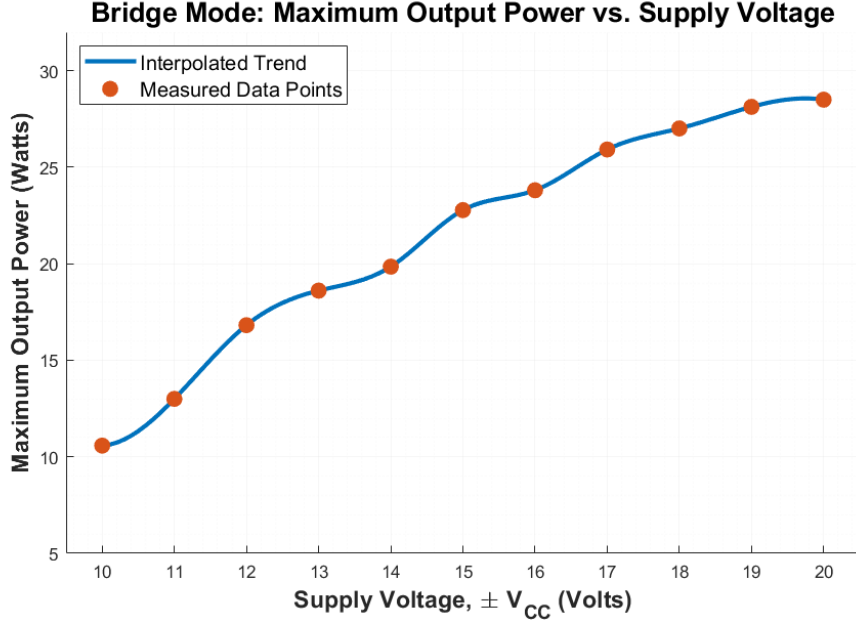


Figure 8: Max Output Power vs Supply Voltage (Bridge)

- The optimal supply voltage range was determined to be around 15V (typical) while satisfying the power requirements for the NE5532 and preventing excessive thermal dissipation in the TIP41/42 pair.
- The test data conclusively verified that the output power **quadruples** when the circuit is switched into Bridge Mode.

5.2.3 Quiescent Current

The quiescent current (I_q) represents the idle current drawn from the power supply when no audio signal is present. It confirms the stability of the Class-AB diode biasing network.

Table 3: Measured Quiescent Current vs. Supply Voltage

Supply Voltage ($\pm V_{CC}$)	Measured I_q (mA)
10 V	16.50
11 V	18.28
12 V	20.11
13 V	21.96
14 V	23.80
15 V	25.66
16 V	27.56
17 V	29.48
18 V	31.37
19 V	33.29
20 V	35.24

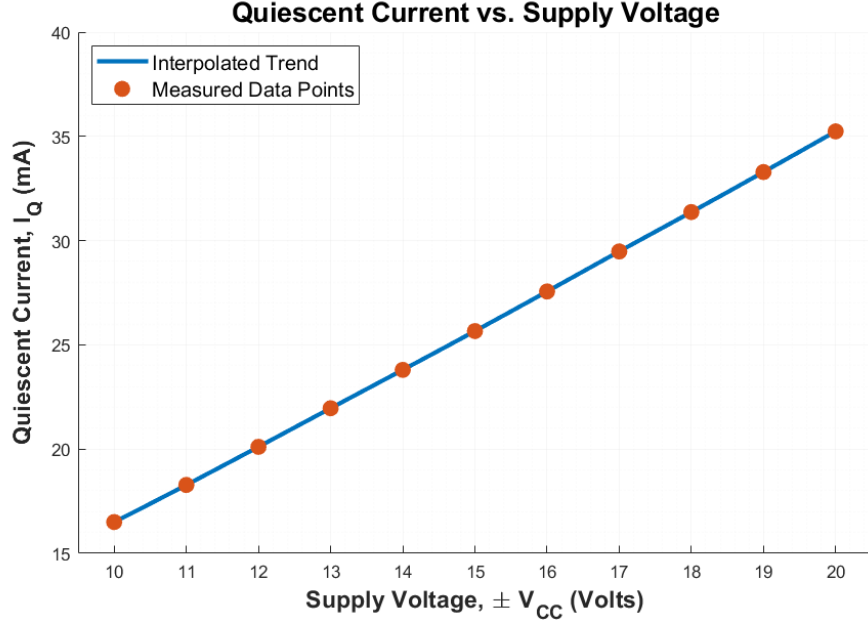


Figure 9: Quiescent Current vs Supply Voltage

The current flowing through the biasing diodes scales linearly with the supply voltage.

5.2.4 Amplifier Efficiency

Efficiency (η) is the ratio of audio power delivered to the load (P_{out}) compared to the total electrical power drawn from the power supply rails (P_{in}). The total power drawn is the sum of the constant wasted idle power (P_{idle}) and the dynamic power drawn while amplifying a sine wave ($P_{dynamic}$):

$$P_{idle} = 2 \cdot V_{CC} \cdot I_q \quad (2)$$

$$P_{dynamic} = \frac{2 \cdot V_{CC} \cdot V_{peak}}{\pi \cdot R_L} \quad (3)$$

$$\eta = \left(\frac{P_{out}}{P_{idle} + P_{dynamic}} \right) \times 100\% \quad (4)$$

Table 4: Calculated Efficiency based on measured Stereo Mode parameters

Supply Voltage ($\pm V_{CC}$)	P_{idle} (W)	$P_{dynamic}$ (W)	P_{out} (W)	Efficiency (η %)
10 V	0.33	7.64	2.88	36.14%
11 V	0.40	8.93	3.25	34.83%
12 V	0.48	11.27	4.35	37.01%
13 V	0.57	12.62	4.65	35.26%
14 V	0.67	14.04	4.96	33.73%
15 V	0.77	15.28	5.12	31.90%
16 V	0.88	17.57	5.95	32.24%
17 V	1.00	19.21	6.30	31.17%
18 V	1.13	20.91	6.66	30.21%
19 V	1.27	23.27	7.41	30.20%
20 V	1.41	24.83	7.61	29.00%

The maximum measured efficiency was around 37%.

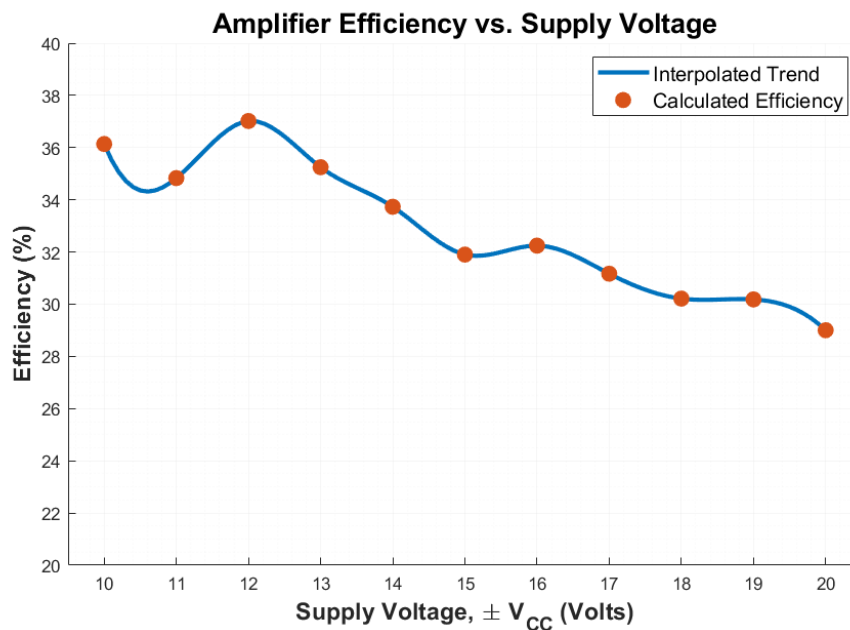


Figure 10: Amplifier Efficiency vs Supply Voltage

5.3 Frequency Response and Harmonics

- Measured Amplifier Bandwidth: 23.8 kHz



Figure 11: Measurement of Slew Rate

- Measured Amplifier Slew Rate: 1.25 V/ μ s (Exceeding the 0.879 V/ μ s requirement).

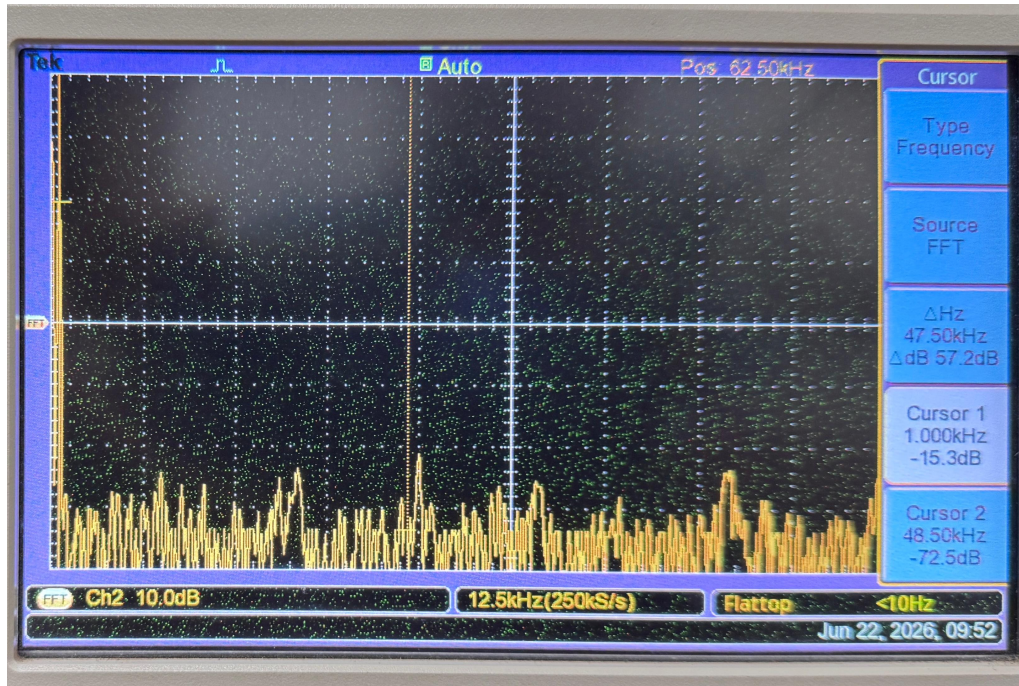


Figure 12: FFT of the output signal given a 1kHz input.

The Fast Fourier Transform (FFT) analysis proved that harmonic distortion from the fundamental 1kHz audio frequency is non-observable, validating the integrity of the core amplifier's design.

References

- [1] Douglas Self. *Audio Power Amplifier Design Handbook*, 4th Edition.
- [2] ALPHA Lab. “How does it work — Amplifier mode — what is the bridge mode in power amplifier.” YouTube, Apr. 14, 2022. [Online]. Available: https://www.youtube.com/watch?v=Yawl4gN_WiQ
- [3] ElectroBOOM. “Making an Audio Power Amplifier (Drive a Sub Woofer).” YouTube, May 31, 2013. [Online]. Available: <https://www.youtube.com/watch?v=06btgjAV3Hk>